

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2199

Chambers Springs 138 kV
February 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2199. The Interconnection Customer proposes a 188.3 MW solar photovoltaic facility interconnecting at the Chambers Springs 138 kV substation in the Pennsylvania area. The request was received on October 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 620 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$6,754,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chambers Springs 345-138 kV XFMR	Vieques 2501 to Geer 138.0 kV	105.8	377	61.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chambers Springs 345/138kV TX	Replace transformer with higher-rated unit	\$23,409,000	29% (\$6,754,000)
	Total allocated to project		\$6,754,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.